

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 1.4: Cell Division — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 11	
Sub-Strand	Sub-Strand 1.4: Cell Division
Driving Question	DRIVING QUESTION: How does the body produce the enormous number of new cells needed for growth?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 From Newborn to Teenager: Where Do All the New Cells Come From?	We watched a time-lapse of a child growing over 14 years and compared baby photographs with how we look now. A newborn of about 3 kg became a teenager of 50 kg or more.	Growth involves a huge increase in the number of cells, and eating food alone does not explain where the new cells come from. We do not yet know how new cells are made or how each gets the right instructions.	Our first ideas are that cells might grow bigger, multiply, or be made from food. We posted questions on the DQB under 'How does growth occur?' and will test our initial models in the next lessons.
Lesson 2 What Is Cell Division? Meaning of Cell Division and the Structure of the Chromosome	We observed diagrams where one cell became two (or more) cells after division, and our observation table recorded the counts. We built a clay model of 1 cell → 2 cells and saw that each new cell needs something from the original cell. We observed an unlabelled chromosome diagram with two identical arms joined at one point.	Cell division is the process by which a parent cell divides to form new daughter cells, which is how the body makes the many new cells needed for growth. Chromosomes carry the genetic information (DNA and genes) that each new cell needs. A chromosome has two chromatids joined at a centromere.	A child grows from 3 kg to 50 kg because existing cells divide to form new cells, not because food turns directly into cells. For the new cells to work properly, the original cell must pass on chromosomes carrying genetic information. We still need to explain what happens to chromosomes during cell division so that each new cell gets the correct genetic information.
Lesson 3 Types of Cell Division: Mitosis and Its Stages	We observed in the mitosis diagram that chromosomes became visible, lined up in the middle of the cell, separated and moved to opposite ends, two new nuclei formed, and the cell divided into two cells.	We learned that mitosis has four stages, prophase, metaphase, anaphase and telophase, followed by cytokinesis, and that it produces two daughter cells with the same chromosomes as the parent cell.	We can explain that a child grows because mitosis copies and shares chromosomes accurately, so one cell becomes two identical cells, which increases cell number for growth and repair. We still ask: if mitosis makes similar body cells, how are the cells for sexual reproduction produced?
Lesson 4 Meiosis and Its Stages:	We observed unlabelled pictures of meiosis and sequenced them, watched videos of	We learned that meiosis is a division in reproductive organs that has two divisions	We can explain that sperm and egg cells must have 23 chromosomes because

Why Do Sperm and Egg Cells Have 23 Chromosomes?	meiosis I and II, read about the stages, and calculated that two cells with 46 chromosomes each would give a zygote with 92 chromosomes.	(meiosis I and meiosis II) and produces four cells, each with half the chromosome number (23). In meiosis I homologous chromosomes separate, and in meiosis II sister chromatids separate.	meiosis halves the chromosome number, so that when they fuse at fertilisation the zygote has 46 chromosomes again and can then divide by mitosis to grow into a child. Without meiosis the number would double in each generation. We still wonder: What would happen to an organism if mitosis and meiosis did not occur?
Lesson 5 Significance of Mitosis and Meiosis: Case Studies from Growth, Healing, Roots and Offspring	Case studies of a growing child, a healing skin cut, a plant producing new roots and shoots, a chicken producing offspring and parents producing children with different characteristics; for each we looked at evidence such as identical or different cells, chromosome numbers and the purpose of the new cells.	Mitosis produces genetically identical diploid cells for growth, repair and asexual reproduction in plants, which is how a child makes billions of new cells with the right instructions. Meiosis produces haploid gametes with different combinations of genes, so fertilisation restores the chromosome number and offspring vary from their parents.	A 3 kg baby becomes a 50 kg teenager because body cells divide repeatedly by mitosis, using materials from food, with each new cell receiving an identical full set of chromosomes. Meiosis is used only to make gametes for sexual reproduction. New question: How can a single organism produce a new organism without producing sex cells?
Lesson 6 Budding and Binary Fission: Making New Cells and Organisms from One Parent	We observed that yeast produces a small bud that grows and separates from the parent, while a bacterium copies its chromosome, elongates and splits into two equal cells. A single bacterium dividing every 20 minutes gives 64 cells after 2 hours.	We learned that budding gives unequal parent and daughter cells (yeast), binary fission gives two equal cells (bacteria), and both involve copying genetic material before the cell divides. Both are fast ways to increase cell numbers.	Budding and binary fission are forms of asexual reproduction in which a new organism arises from a single parent without fusion of sex cells, so the offspring are genetically identical to the parent. This shows how cell division produces new cells, as in the growth of a child, and raises the question of how the mechanisms and types compare in more detail.
Lesson 7 Microscope Practical on Mitosis: Catching Growth in Action in Onion Root Tips	We prepared onion root tip squashes and, using a light microscope, saw cells at different stages of mitosis (interphase, prophase, metaphase, anaphase, telophase) in the growing tip. Most cells were in interphase, and a smaller number were dividing.	Mitosis is how existing cells divide to form two identical daughter cells, each with a full set of chromosomes. Growing tissues have many cells in division. Safety precautions are essential when using microscopes, stains and acids.	A child grows from about 3 kg to 50 kg or more because cells in growing tissues divide by mitosis again and again, doubling the number of cells each time. Food provides the energy and materials, but the new cells come from the division of existing cells, each receiving the same instructions (chromosomes).
Lesson 8 Why Cell Division Matters: From Baby to Teenager, and Bananas from a Test Tube	We observed that growth, repair of injured tissues, and reproduction in organisms such as yeast and plants all involve an increase in the number of cells. We saw how a tissue culture of a banana or tea plant starts with a small piece of plant (explant) that grows into a callus, then shoots and roots, forming many identical plantlets.	We learned that cell division by mitosis produces genetically identical cells for growth, repair and asexual reproduction, while meiosis produces gametes for sexual reproduction and variation. Tissue culture uses repeated mitosis in sterile nutrient medium to produce many disease-free clones quickly. Food provides the raw materials and energy; cell division provides the new cells.	The body produces the enormous number of new cells needed for growth because existing cells divide repeatedly by mitosis, starting from one fertilised egg, doubling the cell number again and again. Each new cell receives a full copy of the chromosomes, so it has the right instructions. The same principle lets farmers in Kenya multiply bananas and tea through tissue culture.

Lesson 9 Consolidation and Digital Animations: Comparing Mitosis and Meiosis to Explain Growth	We watched animations of mitosis and meiosis and saw chromosomes copy, line up, separate and move into new cells. In mitosis one cell made two identical cells with the same number of chromosomes; in meiosis one cell made four cells, each with half the number of chromosomes. We saw models of these stages from our classmates.	Mitosis produces genetically identical diploid cells for growth and repair, such as bone, skin and the root tips of maize. Meiosis produces haploid gametes with different gene combinations for sexual reproduction. DNA is copied before division so that each new cell receives a full set of instructions. Food gives energy and raw materials but does not turn into cells by itself.	A child grows from 3 kg to 50 kg because the fertilised egg divides again and again by mitosis, doubling cell numbers each round (1, 2, 4, 8, 16...) until billions of cells exist. Each new cell receives an exact copy of the chromosomes, so it has the right instructions to work properly. Meiosis earlier formed the egg and sperm that joined to make the first cell, keeping the chromosome number constant from one generation to the next.
Lesson 10 Final Model and Return to the Phenomenon: How Does the Body Make the Billions of New Cells Needed for Growth?	We observed the baby photographs and growth time-lapse again, onion root tip cells in different stages of mitosis, chromosome models, animations of mitosis and meiosis, budding yeast, bacteria dividing by binary fission and examples of tissue culture. We also compared our Lesson 1 drawing with our final model.	We learned that growth happens mainly because the number of cells increases. Existing cells copy their DNA and divide by mitosis into two identical daughter cells with the same chromosome number, so each new cell gets the right instructions. Meiosis makes gametes with half the chromosomes, and fertilisation forms a zygote that begins the growth of a new individual. Budding and binary fission are forms of cell division in yeast and bacteria, and tissue culture uses cell division to grow many plants from small pieces of tissue.	A child grows from 3 kg to 50 kg because the zygote and later body cells divide by mitosis again and again (1, 2, 4, 8, 16...), producing billions of cells. Before each division, DNA is copied, so every daughter cell has identical chromosomes and the right instructions. Food such as ugali, beans and sukuma wiki supplies the raw materials and energy for making new cells, and cells then differentiate into specialised types like muscle, bone, blood and nerve cells.

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